林口長庚內科部腫瘤科病房工作規範

2021/10

壹、科內教學活動

星期	時間	主題	地點
W1-5	07:30-08:30	Morning meeting / 教學住診	6F 討論室
W3	11:30-12:30	Journal Reading / MMI	6F 討論室
依當月排定	依當月排定	病歷迴診教學	6F 討論室
依當月排定	12:00-13:00	醫學倫理暨安寧療護討論會	病房護理站討論室

- 07：30 AM Morning Meeting
  - 報前天白天及值班住院病人(病歷包括舊病歷)
  - 請在 7:30 AM 將病患姓名、年齡、性別及診斷填寫於白板，以免耽誤晨會討論時間及個人考核
  - 內容應包含 Journal Reading、MMI、Case discussion
- 教學住診
  - 依教學部規定，輪訓各專科每週應完成教學住診紀錄表及教學門住診病歷表
- 其他：病歷評核
  - 每月上下半月主治醫師將依各學員所寫之病歷進行線上評核，低於 80 者將提報輔導。
  - 所有學員應依據主治醫師所給評語及分數，進行線上回覆。

癌症團隊會議

癌症團隊	開會時間	開會地點
乳癌	W5 17:00	醫學大樓三樓病理科討論室
頭頸癌	W2 07:30	醫學大樓三樓病理科討論室
腦癌	W2 07:30	復健大樓 3H 精神科會議室
肺癌	W3 16:00	醫學大樓一樓胸腔科第 16 號門討論室
肝癌	W3 17:00	病理大樓一樓 X 光科圖書室
甲狀腺癌	一、三週之 W1 14:00	醫學大樓三樓病理科討論室
食道癌	W3 16:00	醫學大樓三樓病理科討論室
胃癌	W4 16:30	醫學大樓 5C 會議室
大腸直腸癌	W2 07:10	病理大樓一樓 X 光科圖書室
消化道癌(腸胃道基質瘤、膽道癌、胰臟癌)	W4 17:00	醫學大樓 5C 會議室
皮膚癌	二、四週之 W5 12:30	兒童大樓 L 棟 B1 皮膚科討論室
泌尿道癌	W2 07:30	復健大樓 13G 會議室
軟組織肉瘤/骨癌	二、四週之 W4 13:00	復健大樓施信農醫師門診
免疫治療	一、三週之 W3 14:30	醫學大樓三樓病理科討論室
精準醫療	每月最後一週 W3 14:30	地點不定

主治醫師	參加之癌症團隊
王宏銘	頭頸癌
楊再勝	肝癌、大腸直腸癌
林永昌	乳癌、泌尿道癌
陳仁熙	胃癌、消化道癌(腸胃道基質瘤、膽道癌、胰臟癌)
張献崑	乳癌、食道癌
張文震	免疫治療、精準醫療、肺癌、皮膚癌
廖繼鼎	軟組織肉瘤/骨癌、腦癌
徐正龍	頭頸癌、鼻咽癌(接頭頸癌團隊會議後)
沈雯琪	消化道癌(腸胃道基質瘤、膽道癌、胰臟癌)、乳癌
徐鴻智	大腸直腸癌
侯明模	免疫治療、精準醫療、食道癌、肝癌、胃癌、消化道癌(腸胃道基質瘤、膽道癌、胰臟癌)
周文其	胃癌、消化道癌(腸胃道基質瘤、膽道癌、胰臟癌)
蘇柏榮	食道癌、乳癌、泌尿道癌、消化道癌(腸胃道基質瘤、膽道癌、胰臟癌)

吳教恩	食道癌、肺癌、消化道癌(腸胃道基質瘤、膽道癌、胰臟癌)、甲狀腺癌、皮膚癌
黃文冠	泌尿道癌、消化道癌(腸胃道基質瘤、膽道癌、胰臟癌)
彭夢婷	食道癌、免疫治療、精準醫療、乳癌
黃姵瑋	頭頸癌、消化道癌(腸胃道基質瘤、膽道癌、胰臟癌)、乳癌、大腸直腸癌
張境夫	大腸直腸癌、肺癌、消化道癌(腸胃道基質瘤、膽道癌、胰臟癌)
余紹銘	大腸直腸癌、泌尿道癌、消化道癌
徐執中	食道癌、肝癌、乳癌

* 請各 Clerk 跟隨指導主治醫師參加癌症團隊會議

貳、每日常規工作：

1、 每日查新住院病人，看病人處理 complaints。progression note 以 SOAP 的形式書寫，每日需紀錄疼痛評估、performance status 及相關 physical examination，病例請詳細填寫 diagnosis、treatment、response(CR、PR、SD、PD)，toxicity grading and other complications 及主治醫師查房時的 comment，善加利用 TPR sheet 記錄 special procedure or antibiotics use。每週寫一次 weekly summary，並包含 measurable goal。
2、 請主動查 data，追蹤各種檢查結果（如 X-ray、CT、pathology、會診）。
3、 開任何 order 都要開單，如檢驗單、藥單。請全部用電腦開單。
4、 請每週 order renew (medication + IVF renew)。
5、 實習醫師開 order，住院醫師要記得 counter sign。
6、 Patient Controlled Analgesia（PCA）各 station 有固定 order form 可用。
7、 實習醫師或住院醫師須視病情需要，主動參與換藥，不要一切推諉於護理人員。若無需換藥，請 DC order。
8、Invasive procedure (eg. chest tube insertion, CVP, pleurocentesis, ascites tapping)，請依照醫師職權規範執行，如職權內仍無法順利完成，請知會 Fellow 處理。
9、有 Oncology Emergency case 要立即處理，並通知 Fellow and VS；CPR 務必通知病房 CR and VS。
10、當日預出院病患請儘量於中午 12 點以前辦理出院，明日預出院可辦理「今辦明出」。
11、New patient 於下午 4：00 以後原則上由值班醫師接，處理原照顧病人至 PM 4：30 交班。
12、裝完 port -A 後需 Follow CXR。位置正確者，無 pneumothorax 請於病歷上註明【port- A 可用】及【port-A irrigation ×次數】。
13、出院病歷摘要需註明病患出院情況、出院帶藥、回診日期。
14、值班時由住院醫師決定 UGY 醫師接哪兩個 new patient。無法處理的 case，務必通知上級醫師。
15、請遵守值班室各項規定，並留下 GSM 與值班室電話。若值班時屢 call 不到者，視同曠職及未值班。
16、病患請假需由主治醫師同意，值班時找 duty CR。
17、開會或上課時，病人若有緊急情況發生，須優先處理。
18、若化學治療室 OPD 病患需要做 procedure 時，VS 會請其住院醫師 cover。
19、高血鈣處理要注意最近是否已使用過雙磷酸鹽類藥物或 denosumab，如果血鈣濃度已下降，應避免短期內再度使用同類藥物，後續需適時監測血鈣濃度。

關於 chemotherapy：

1、化療 order 需由主治醫師或專科醫師開立或覆核。化學藥物調劑單要開清楚，並開立化療處置單。
2、住院醫師依據 data 來決定病人是否接受化療，並執行化學藥物的注射。
3、請儘早開立 chemotherapy order，以減少值班人員的 loading。

關於電子交接班：

1、每日務必點選「電子交班」，並以 ISBAR 的形式填寫，如不需交班的病人請點“不需交班”，有 critical case 請主動跟病房 CR 報備，下班前要與當天 duty resident 交班，以免造成值班醫師困擾。

2、符合以下任一條件之病患，醫師須交接班：

— 病危病患
— 符合「本院臨床警訊呼叫標準」之病人
— 當天由 ICU 轉出之病患
— 侵入性檢查治療後有併發症之病患
— 接受化學治療發生非預期之嚴重病發症病患

本院臨床警訊呼叫標準

● Respiratory Rate	≦8 或≧29
● Pulse（bpm）	≦45 或≧120
● SpO2	<90%
● Systolic BP（mmHg）	<90mmHg 或>200mmHg 且經治療後無法改善
● 神經症狀	發生意識不清情況
	Seizures 或發生任何不正常之肢體動作
	胸痛且臉色蒼白及盜汗
● 其他	每班尿量<200c.c. (排除規則洗腎病患)
	如果醫師/護師判定需要啟動者

＊如果病人新出現上述任一徵候，立即通知醫師。

參、PUSH 化學藥物注意事項：

化學治療室中，push chemotherapy 的工作由各 VS 之實習醫師或住院醫師負責完成

1、 應先確定病人姓名及藥物，戴好口罩、手套。
2、 自 Peripheral line 注入，要確定有回血才開始 slow push，否則 reset new line。
自 port-A 注入，要確定有回血，否則先 run NS 300cc 若沒有腫脹情形才開始 slowly push。自 port-A 注射藥物理論上不應該有疼痛情形。
3、 若病人有疼痛情形即使有回血應該懷疑藥物外漏，要馬上停止注射，立即儘量回抽藥物，再依藥物特性處理，並告知 CR、VS，並於病歷上記錄。
4、 完成化學藥物 push 後請將針筒放入塑膠袋中依毒性廢棄物處理。
5、 應隨時觀察 push chemotherapy 後的反應，以便早期發現 side effects or complications。

肆、腫瘤科臨床六大臨床核心能力訓練範例

(一) 癌症疼痛【Cancer Pain】

癌症疼痛 Cancer Pain 是腫瘤科病人常見的合併症，如何處理好癌症病患的疼痛，是個很重要的學習課程。

＊學習目標＊

一、 醫學知識 *(MK)*：實習醫師學員需認識、了解、討論下面議題—

1. 癌症疼痛的原因 *(MK)*

(1)癌症的浸潤
(2)治療(手術、放射線治療、化學治療等)導致
(3)非癌症疼痛 (Non-Cancer pain) *(MK)*:

- 黏膜發炎 (Mucositis)
- 肌肉及軟組織疼痛 (Muscles and connective tissue)

2. 癌症疼痛的種類 *(MK)*:
    - 體感性疼痛 (Somatic pain)
    - 內臟性疼痛 (Visceral pain)
    - 神經病變性疼痛 (Neuropathic pain)
3. 癌症疼痛的型態 *(MK)*:
    - 持續性的疼痛 (persistent pain)
    - 突發性的疼痛 (breakthrough pain)
4. 複雜性的癌症疼痛 *(MK)*
    - 癌症疼痛可由癌症本身、癌症的合併症及治療的副作用等導致
    - 癌症疼痛可能同時存在不同種類的疼痛
    - 了解各類癌症疼痛的原因、位置、性質、程度及相關症狀是相當重要的
5. 癌症疼痛控制的原則 *(MK)*
    - 按 WHO 三階梯(three-step)的疼痛控制進行
    - 疾病過程須含括身體及精神的照料 (physical & psychological modalities)
    - 對原來疾病做適當的治療 (appropriate primary therapy)
    - 如疼痛控制失敗，可藉助疼痛科或神經外科醫師，用神經調製或神經破壞的方法 (neuromodulative & neurodestructive procedures) 來改善頑強性的疼痛

二、 病患照護與臨床技能*(PC, CS)*: UGY 學員需具備下列技能

1. 病史詢問: 學員須有獲得、確認、並能報告完整的癌症病患疼痛的病史－
    - 癌症疼痛的原因 *(PC, CS)*
    - 癌症疼痛的位置 *(PC, CS)*
    - 癌症疼痛的性質 *(PC, CS)*
    - 癌症疼痛的程度 *(PC, CS)*
    - 癌症疼痛的相關症狀 *(PC, CS)*
2. 從理學檢查得到診斷與評估嚴重度:
    - 正確的從理學檢查測出疼痛的位置 *(PC, CS)*
    - 正確的從理學檢查測出與疼痛相關的神經症狀 *(PC, CS)*
    - 正確的從理學檢查測出與疼痛相關的其他症狀 *(PC, CS)*
    - 嚴重度
    - 疼痛嚴重度的評估 *(PC, CS)*
      輕度 (1-4) 、中度 (5-7) 、輕度 (8-10) 、
    - 癌症疼痛的評估表 *(PC, CS)*
      視覺模擬評分法 Visual analogue scales (VAS)
      數字評價量表 Numerical rated scales (NRS)
      詞語等級評分法 Verbal rated scale (VRS)
3. 鑑別診斷: 學員能從癌症疼痛的特殊病史與理學檢查中:
    - 做出癌症疼痛的特質 *(PC, MK)*
    - 分辨出癌症疼痛的位置 *(PC, MK)*
    - 分辨出癌症疼痛的種類 *(PC, MK)*
    - 分辨出癌症疼痛，包括:
      神經根 (nerve root) *(PC, MK)*
      神經節 (nerve ganglion) *(PC, MK)*
      神經叢 (nerve plexus) *(PC, MK)*
      周邊神經(peripheral nerve) *(PC, MK)*
4. 數據判讀: 學員能判讀癌症疼痛常用之影像檢查:

可能做的診斷性的影像檢查應包括:

- 一般 X 光檢查 (plain film) *(PC, MK)*
- 電腦斷層攝影檢查 (CT scan) *(PC, MK)*
- 核磁共振檢查 (MRI) *(PC, MK)*
- 核子醫學影像檢查 (radionuclide, 如骨骼掃描及正子檢查) *(PC, MK)*

5. 人際與溝通技巧 (ICS): 學員須能－
   - 與病患及家屬溝通討論癌症疼痛的診斷、治療與追蹤之計畫 *(PC, CS)*
   - 衛教病患與癌症疼痛相關之個人與環境的因素*(PC, CS)*
6. 病患處置: 學員能建立一套癌症疼痛處置之流程計畫

了解疼痛藥物之途徑、劑量、對等量、禁忌、作用機轉、交互作用、副作用與成本效應。**須先親自評估過病人後再開立藥物，若病人從未接受過管制止痛藥(例如: 可待因、嗎啡)，應向病人及家屬解釋原因及可能發生的副作用，同意之下再開立藥物，減少病人及家屬的疑慮及恐慌**。

- 藥物處置
  - 非鴉片類的止痛劑 (Non-Opioid Analgesics) *(PC, MK)*
    Acetaminophen、NSAIDs
  - 鴉片類的止痛劑 (Opioid Analgesics) *(PC, MK)*
    Tramadol、Ultracet、Morphine、MST、Transdermal Fentanyl
  - 鴉片類的止痛劑的對等量 *(PC, MK)*
  - 鴉片類的止痛劑的使用原則 *(PC, MK)*
  - 突發痛(Breakthrough pain) 的處理原則 *(PC, MK)*
  - 自控式止痛裝置 Patient-Controlled Analgesia (PCA) 的使用原則 *(PC, MK)*
  - 輔助性止痛劑(Adjuvant Analgesics)
    Corticosteroids (Dexamethasone)、Antidepressants、Anticonvulsants
- 非藥物處置
  - 精神學方面的治療 (psychologic therapy) *(PC, MK)*
  - 復健方面的技術 (physiatric technique) *(PC, MK)*
- 頑強性癌症疼痛的處置
  - 疼痛科及神經外科方面(anaesthesic and neurosurgical approach) *(PC, MK)*
    a) 緩解腫瘤對神經的壓迫 (relief of neural compression by tumor) *(PC, MK)*
    b) 藥物直接注入中樞神經系統 (delivery of drugs directly to CNS) *(PC, MK)*
    c) 用電刺激來增強疼痛調控路徑 (augmentation of pain-controlling pathways by electric stimulation) *(PC, MK)*
    d) 阻斷疼痛路徑 (interruption of pain pathways) *(PC, MK)*
- 癌症末期病人照顧的理念
  a) 繼續使用鴉片類的止痛劑但不須減量。使他們死的舒服 *(PC, MK)*
  b) 使癌症末期病人能鎮定(Sedation) *(PC, MK)*

三、 執業中學習與改進:

從病患、主治醫師的教學與經驗、書刊與網路尋找癌症疼痛相關議題之資料

四、 態度與專業行為: 學員能力

1. 提出合乎實證醫學、成本效益及風險利益，評估癌症疼痛的影像檢查及治療 (PLI, P),
2. 認識癌症疼痛對生活品質的影響 (P),
3. 自我學習癌症疼痛的相關議題 (PLI, P),

(二)【癌症惡病體質 Cancer Cachexia】

癌症惡病體質 Cancer Cachexia 是腫瘤科病人易見的癌症伴隨症侯群,如何診斷及處理這些癌症病患，是個很重要的學習課程。

＊學習目標＊

一、 醫學知識 *(MK)*: 實習醫師學員需認識、了解、討論下面議題－

1. 癌症惡病體質的定義 *(MK)*:

(1)厭食 (anorexia)、胃口差 (poor appetite)

(2)體重明顯的下降 (>5 %體重減輕/3-6 個月)

(3)肌肉萎縮、身體體能狀況變差

2. 癌症惡病體質造成的體重明顯的下降 *(MK)*:

- 以胃腸癌症，尤其胰臟癌及胃癌最為明顯
- 癌症惡病體質者比非癌症惡病體質者，化學治療的療效差
- 癌症惡病體質者比非癌症惡病體質者，死亡率高
- 癌症惡病體質者 30%以上的體重下降，常會致命

3. 癌症惡病體質的原因 *(MK)*:

- 飲食攝取變少
- 身體的代謝增加

➢ 癌症惡病體質病患的代謝性變化. (metabolic changes in cancer patients) *(MK)*:

a) 醣類的代謝(carbohydrate metabolism)

- 增加 Cori 循環的活動度 (increase Cori cycle activity)
- 胰島素抗阻 (insulin resistance)

b) 脂肪的代謝 (fat metabolism)

- 加強脂肪的動化 (enhanced lipid mobilization)
- 降低脂肪蛋白脂肪脢 (decreased lipoprotein lipase)

c) 蛋白質的代謝 (protein metabolism)

- 降低肌肉蛋白的形成(decreased muscle protein synthesis)
- 增加肌肉蛋白的代謝 (increased muscle catabolism)
- 肝臟蛋白的產生 (increased liver protein synthesis)

－增加急性期蛋白的產生(increased acute phase protein synthesis)

➢ 癌症惡病體質的代謝中間物. (catabolic mediators)

－ 分腫瘤及體液 (tumoral & humoral)兩方面 *(MK)*:

- 腫瘤或病患本身藉著免疫細胞產生細胞素(cytokines): *(MK)*

  Cytokines 包括:

  Tumor-necrotic factor- alpha (TNF-α)、Interleukin-1 (IL-1)、IL-6、IL-8、Interferon-gamma (INF-r)、Leptin、Vascular endothelial growth factor- alpha (VEGF--α)
- 腫瘤產生的物質:

  脂肪動員因素 (lipid-mobilizing factors)

  蛋白質動員因素 (protein-mobilizing factors

  蛋白溶解誘發因素 (proteolysis-inducing factor)

➢ 癌症惡病體質最後導致肌肉質量的大量損失(loss of skeletal muscle) *(MK)*:

－ 藉著 Cytokines 及 protein-mobilizing factors,活化泛蛋白酶體蛋白蛋白溶解系統 (ubiquitin-proteasome proteolytic system) ，造成肌肉蛋白高度的代謝

➢ Cytokines 對人體的影響: *(MK)*:

- 作用於腦造成厭食
- 作用於脂肪組織造成脂肪的代謝增加
- 作用於蛋白質造成周邊肌肉的代謝增加
- 作用於肝臟造成增加急性期蛋白的產生
- 誘發氧化壓力(Oxidative stress) 造成細胞破壞

二、 病患照護與臨床技能*(PC, CS)*: UGY 學員需具備下列技能

1. 病史詢問: 學員須有獲得、確認、並能報告完整的癌症病患惡病體質的病史
    - 病患的食慾方面，是否變差 *(PC, CS)*
    - 病患的體重減輕的情形，是否明顯下降*(PC, CS)*
    - 病患的體能狀況，是否容易疲憊及厭倦 *(PC, CS)*
    - 病患的行動情形，是否覺得無力*(PC, CS)*
    - 病患的睡眠型態，是否發生了改變. *(PC, CS)*
    - 病患的社交活動，是否受到影響. *(PC, CS)*
2. 從理學檢查得到診斷與評估嚴重度:
    - 病患的整個外觀及意識狀況 *(PC, CS)*
    - 病患的體重變化 *(PC, CS)*
    - 病患的四肢骨骼肌的肌肉強度，是否已發生了萎縮
    - 病患的皮膚脂肪及質地 *(PC, CS)*
3. 鑑別診斷: 學員能從癌症病患惡病體質的特殊病史與理學檢查中:
    - 體重明顯的下降是否因由其他內科疾病 *(PC, MK)*
    - 體重明顯的下降、胃口差、睡眠差是否為精神科疾病 *(PC, MK)*
    - 體重明顯的下降是否只因為攝取不足 *(PC, MK)*
4. 數據判讀: 學員能判讀癌症惡病體質常用之檢查及實驗項目，可能做的診斷包括:
    - 基本的血液及生化檢查,包括血色素、蛋白質、血鈣及 C-reactive protein 等 *(PC, MK)*
    - 影像檢查了解腫瘤的狀況 *(PC, MK)*
5. 人際與溝通技巧 *(ICS)*: 學員須能－
    - 與病患及家屬溝通討論癌症惡病體質的診斷、治療與追蹤之計畫 *(PC, CS)*
    - 衛教癌症病患惡病體質的相關照顧*(PC, CS)*
6. 病患處置: 學員能建立一套癌症惡病體質處置之流程計畫
    - 了解癌症病患惡病體質可給的藥物、藥的作用機轉、副作用與成本效應 *(PC, CS)*
    - 了解癌症病患惡病體質是否可藉補給營養來改善、補給營養的副作用與成本效應 *(PC, CS)*
    - 了解是否有其他治療方法來改善癌症惡病體質; 或者已進入癌症末期的安寧緩和照顧 *(PC, CS)*

    ➢ 藥物處置的目的:

    - 讓病患了解癌症惡病體質僅是穩定及維持現況,要治癒或回復原來的身體狀況很難 *(PC, CS)*
    - 癌症惡病體質是改善生活品質，對大多數的病患僅是改善胃口及增加食物的攝取 *(PC, CS)*

    ➢ 藥物處置

    - 藥物用來增加食物的攝取及改善厭食(agents that increase food intake and alleviate anorexia) *(PC, MK)*
        a. Metoclopramide
        b. Megestrol acetate, Medroxyprogesterone acetate
        c. Steroids (Dexamethasone, Methylprednisolone)

- 藥物用來回復代謝的效應 (agents that reversing metabolic effects) *(PC, MK)*
  a. Eicosapentaenoic acid (EPA)
  b. Branched chain amino acids

— 以 Megestrol acetate 及 Medroxyprogesterone acetate 最常使用

○ 營養補給

限於高選擇性的癌症惡病體質病患 *(PC, MK)*

體能狀況差者及存活差的病患不適宜 *(PC, MK)*

三、 執業中學習與改進:

從病患、主治醫師的教學與經驗、書刊與網路尋找癌症惡病體質相關議題之資料

四、 態度與專業行為: 學員能力

1. 提出合乎實證醫學、成本效益及風險利益各方評估癌症惡病體質的診斷及治療 *(PLI, P),*
2. 認識癌症惡病體質的病患，生活方面造成的影響 *(P),*
3. 自我學習癌症惡病體質的相關議題 *(PLI, P),*

備註：

1. *MK — Medical knowledge*
2. *PC —Patient care*
3. *CS — Clinical skills*
4. *ICS — Interpersonal; and communication skills*
5. *P — Professionalism*
6. *SBP —System-based practice*
7. *PCI —Practice-based learning and improvement*

伍、ECOG performance statue scale

Grade	Level of activity
0	Fully active, able to carry on all predisease performance without restriction (Karonfsky 90-100%)
1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature. e.g., light house work, office work (Karnofsky 70-80%)
2	Ambulatory and capable of all self-care but unable to carry out any work activities; up and about more than 50% of waking hours (Karnofsky 50-60%)
3	Capable of only limited self-care, confined to bed or chair more than 50% of waking hours (Karnofsky 30-40%)
4	Completely disable; cannot carry on any self-care; totally confined to bed or chair (Karnofsky 10-20%)

陸、Grading of acute and subacute adverse events (NCI CTCAE V5.0)

■ **Common adverse events**

	CTCAE Term	Grade 1	Grade 2	Grade 3	Grade 4	Grade 5
Hematologic	Anemia	Hemoglobin (Hgb) <LLN - 10.0 g/dL	Hgb <10.0 - 8.0 g/dL	Hgb <8.0 g/dL	Life-threatening consequences; urgent intervention indicated	Death
	White blood cell decreased	<LLN - 3000/mm3	<3000 - 2000/mm3	<2000 - 1000/mm3	<1000/mm3; <1.0 x 10e9 /L	-

	Neutrophil count decreased	<LLN - 1500/mm3	<1500 - 1000/mm3	<1000 - 500/mm3	<500/mm3; <0.5 x 10e9 /L	-
	Febrile neutropenia	-	-	ANC <1000/mm3 with a single temperature of >38.3 degrees C (101 degrees F) or a sustained temperature of >=38 degrees C (100.4 degrees F) for more than one hour	Life-threatening consequences; urgent intervention indicated	Death
	Platelet count decreased	<LLN - 75,000/mm3	<75,000 - 50,000/mm3	<50,000 - 25,000/mm3; <50.0 - 25.0 x 10e9 /L	<25,000/mm3; <25.0 x 10e9 /L	-
Gastrointestinal	Nausea	Loss of appetite without alteration in eating habits	Oral intake decreased without significant weight loss, dehydration or malnutrition	Inadequate oral caloric or fluid intake; tube feeding, TPN, or hospitalization indicated	-	-
	Vomiting	Intervention not indicated	Outpatient IV hydration; medical intervention indicated	Tube feeding, TPN, or hospitalization indicated	Life-threatening consequences	Death
	Mucositis oral	Asymptomatic or mild symptoms; intervention not indicated	Moderate pain or ulcer that does not interfere with oral intake; modified diet indicated	Severe pain; interfering with oral intake	Life-threatening consequences; urgent intervention indicated	Death
	Diarrhea	Increase of <4 stools per day over baseline; mild increase in ostomy output compared to baseline	Increase of 4 - 6 stools per day over baseline; moderate increase in ostomy output compared to baseline; limiting instrumental ADL	Increase of >=7 stools per day over baseline; hospitalization indicated; severe increase in ostomy output compared to baseline; limiting self care ADL	Life-threatening consequences; urgent intervention indicated	Death
	Enterocolitis	Asymptomatic Clinical or diagnostic observations only; intervention not indicated	Abdominal pain; mucus or blood in stool	Severe or persistent abdominal pain; fever; ileus; peritoneal signs	Life-threatening consequences; urgent intervention indicated	Death
	Alanine aminotransferase increased	>ULN - 3.0 x ULN if baseline was normal; 1.5 - 3.0 x baseline if baseline was abnormal	>3.0 - 5.0 x ULN if baseline was normal; >3.0 - 5.0 x baseline if baseline was abnormal	>5.0 - 20.0 x ULN if baseline was normal; >5.0 - 20.0 x baseline if baseline was abnormal	>20.0 x ULN if baseline was normal; >20.0 x baseline if baseline was abnormal	-

	Aspartate aminotransferase increased	>ULN - 3.0 x ULN if baseline was normal 1.5 - 3.0 x baseline if baseline was abnormal	>3.0 - 5.0 x ULN if baseline was normal; >3.0 - 5.0 x baseline if baseline was abnormal	>5.0 - 20.0 x ULN if baseline was normal; >5.0 - 20.0 x baseline if baseline was abnormal	>20.0 x ULN if baseline was normal; >20.0 x baseline if baseline was abnormal	-
	Blood bilirubin increased	>ULN - 1.5 x ULN if baseline was normal > 1.0 - 1.5 x baseline if baseline was abnormal	>1.5 - 3.0 x ULN if baseline was normal; >1.5 - 3.0 x baseline if baseline was abnormal	>3.0 - 10.0 x ULN if baseline was normal; >3.0 - 10.0 x baseline if baseline was abnormal	>10.0 x ULN if baseline was normal; >10.0 x baseline if baseline was abnormal	-
Skin and subcutaneous tissue	Alopecia	Hair loss of <50% of normal for that individual that is not obvious from a distance but only on close inspection; a different hair style may be required to cover the hair loss but it does not require a wig or hair piece to camouflage	Hair loss of >=50% normal for that individual that is readily apparent to others; a wig or hair piece is necessary if the patient desires to completely camouflage the hair loss; associated with psychosocial impact	-	-	-
	Rash acneiform	Papules and/or pustules covering <10% BSA, which may or may not be associated with symptoms of pruritus or tenderness	Papules and/or pustules covering 10 - 30% BSA, which may or may not be associated with symptoms of pruritus or tenderness; associated with psychosocial impact; limiting instrumental ADL; papules and/or pustules covering > 30% BSA with or without mild symptoms	Papules and/or pustules covering >30% BSA with moderate or severe symptoms; limiting self-care ADL; associated with local superinfection with oral antibiotics indicated	Life-threatening consequences; papules and/or pustules covering any % BSA, which may or may not be associated with symptoms of pruritus or tenderness and are associated with extensive superinfection with IV antibiotics indicated	Death
	Rash maculo-pap	Macules/papules covering <10%	Macules/papules covering 10 - 30% BSA with or without	Macules/papules covering >30% BSA with moderate or	-	-

	ular	BSA with or without symptoms (e.g., pruritus, burning, tightness)	symptoms (e.g., pruritus, burning, tightness); limiting instrumental ADL; rash covering > 30% BSA with or without mild symptoms	severe symptoms; limiting self care ADL		
Others	Creatinine increased	>ULN - 1.5 x ULN	>1.5 - 3.0 x baseline; >1.5 - 3.0 x ULN	>3.0 x baseline; >3.0 - 6.0 x ULN	>6.0 x ULN	-
	Ejection fraction decreased	-	Resting ejection fraction (EF) 50 - 40%; 10 - 19% drop from baseline	Resting ejection fraction (EF) 39 - 20%; >=20% drop from baseline	Resting ejection fraction (EF) <20%	-

***其他詳見 https://evs.nci.nih.gov/ftp1/CTCAE/About.html ***

柒、Chemotherapy-induced nausea and vomiting (CINV)

- **Types of emesis**
  - **Acute emesis:** emesis occurring during the first 24 hours after chemotherapy
  - **Delayed emesis:** emesis occurring more than 24 hours after chemotherapy
  - **Anticipatory emesis:** a conditioned response in patients who have developed significant nausea and vomiting during previous cycles of chemotherapy
- **Mechanism**
  - A direct interaction with the AP, induced by antineoplastic agents
  - 5-HT from the enterochromaffin cells (damaged by chemotherapy)
    → 5-HT3 receptors on vagal afferent terminals in the wall of the bowel
    → projecting to the CTZ of dorsal brain stem
    chemoreceptor trigger zone, CTZ = dorsal vagal complex
    = nucleus tractus solitarius (NTS) + area postrema (AP)
  - Substance P, released from sensory neurons following chemotherapy administration
    → act on abundant NK1 receptors within CTZ
- **Classifications of anti-emetic agents**
  - NK1 receptor antagonist: aprepitant (Emed)
  - 5-HT3 receptor antagonist: granisetron (Zofran), ondansetron (Kytril), palonosetron (Aloxei)
  - Dexamethasone
  - Lower therapeutic index: Metoclopramide, Butyrophenones, Phenothiazines, and Cannabinoids
  - Adjunctive drugs: Lorazepam and diphenhydramine
- **Summary of Recommendations for Antiemetics in Oncology：Antiemetic Regimens**

High emetic risk, >90%	
NK1 antagonist	Aprepitant 125 mg oral on D1; 80 mg oral on D2 &D3
	Fosaprepitant 150 mg IV on D1
5-HT3 antagonist (on D1 only)	Granisetron 2 mg oral; 1 mg or 0.01mg/kg IV
	Ondansetron 8 mg oral twice daily; 8mg or 0.15 mg/kg IV
	Palonosetron 0.50 mg oral; 0.25 mg IV
	Dolasetron 100 mg oral ONLY

	Tropisetron 5 mg oral; 5 mg IV
	Ramosetron 0.3 mg IV
Corticosteroid†	Dexamethasone 12 mg oral/IV on D1; 8 mg oral/IV on D2~D4
Moderate emetic risk, 30% to 90%	
5-HT3 antagonist	Palonosetron 0.50 mg oral or 0.25 mg IV on D1 Or Granisetron, Ondansetron (1st 5-HT antagonist)
Corticosteroid	Dexamethasone 8 mg oral/IV on D1, D2 & D3
Low emetic risk, 10% to 30%	
Corticosteroid	Dexamethasone 8 mg oral/IV on D1

■ **Classifications of chemotherapy regimen in different emogenic risks**

High emetic risks	Carmustine Cisplatin AC regimen Cyclophosphamide ≥ 1,500 mg/m2	Dacarbazine Dactinomycin Mechlorethamine Streptozotocin
Moderate emetic risks	Azacitidine Alemtuzumab Bendamustine Carboplatin Clofarabine Cyclophosphamide < 1,500 mg/m2 Cytarabine > 1,000 mg/m2	Daunorubicin Doxorubicin Epirubicin Idarubicin Ifosfamide Irinotecan Oxaliplatin
Low emetic risks	Fluorouracil Bortezomib Cabazitaxel Catumaxomab Cytarabine ≤ 1,000 mg/m2 Docetaxel Doxorubicin HCL liposome injection Etoposide Gemcitabine	Ixabepilone Methotrexate Mitomycin Mitoxantrone Paclitaxel Panitumumab Pemetrexed Temsirolimus Topotecan Trastuzumab
Minimal emetic risks	Chlorodeoxyadenosine Bevacizumab Bleomycin Busulfan Cetuximab	ludarabine Pralatrexate Rituximab Vinblastine Vincristine Vinorelbine

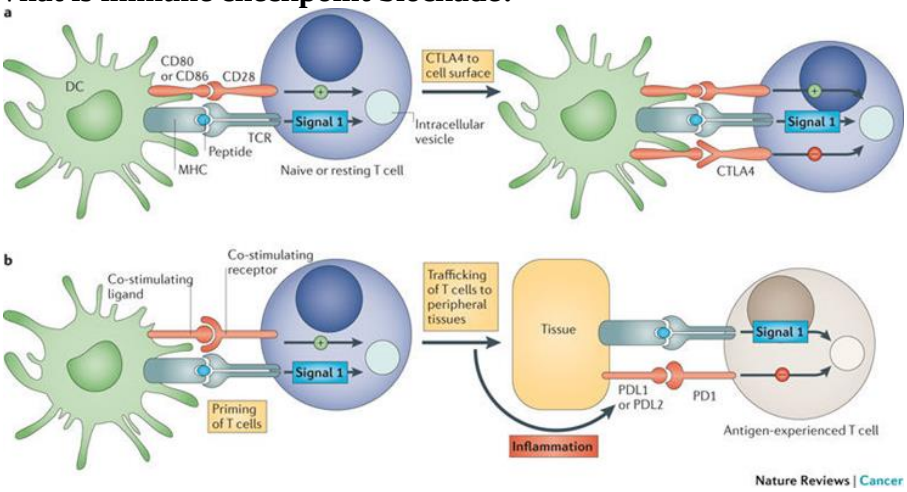
■ **Reference**

UpToDate: Prevention and treatment of chemotherapy-induced nausea and vomiting

Antiemetics: American Society of Clinical Oncology Clinical Practice Guideline Update, *J Clin Oncol 29:4189-4198*

捌、免疫治療相關不良反應 (Immune-related adverse events, irAEs)

■ **What is immune checkpoint blockade?**



Checkpoint inhibitors are a type of immunotherapy. They block proteins that stop the immune system from attacking the cancer cells. Immune checkpoint blockade will inhibit the inhibitory signal to the T cell then start to kill cancer cells.

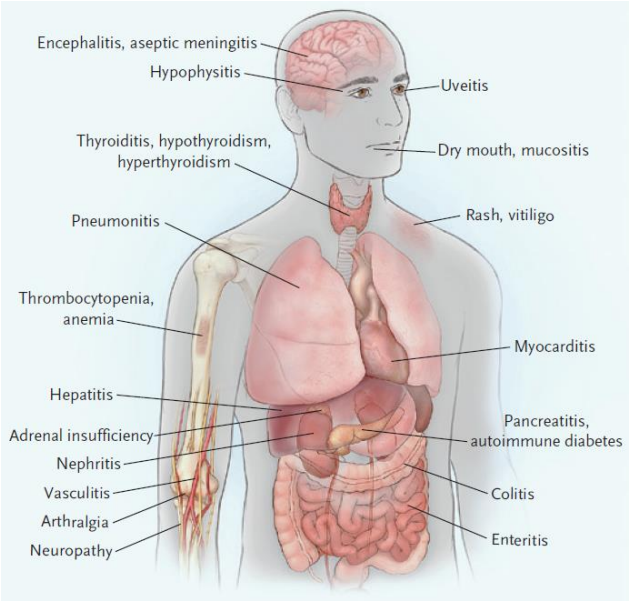
■ **Available immune checkpoint inhibitors in Taiwan (2018.06)**

Drug	Trade name	Target
Ipilimumab	YERVOY®	CTLA-4
Nivolumab	OPDIVO®	PD-1
Pembrolizumab	KEYTRUDA®	PD-1
Atezolizumab	TECENTRIQ®	PD-L1

■ **Adverse events related to immune checkpoint inhibitors**

By increasing the activity of the immune system, immune checkpoint blockade can have inflammatory side effects, which are often termed immune-related adverse events.

- **Any organ or tissue could be affected.**
  Most commonly involved organ:
  → Endocrine: Thyroiditis, hypophysitis, adrenal insufficiency, DM
  → Lung: Pneumonitis
  → Gastrointestinal tract: Colitis,
  → Liver: hepatitis
  → Skin: Rash, vitiligo



■ **Recommendation of survey for irAEs (2018.06)**

- Once patient has clinical symptom, we should always consider the possibility of irAEs. Complete history taking, PE, laboratory data, image exam, biopsy (if could) and rule out other possible common disease are necessary to make a diagnose of irAEs.
- Evaluate severity (Please refer to CTCAE grading system)
- In most of case (especially grade 3 or 4)

→ Informed Chief resident or attending ASAP!
→ Discontinue Immunotherapy
→ Glucocorticoid should be given right away unless contraindicated. (start with 0.5-2mg/kg/day methylprednisolone, NCCN or ESMO guideline)
→ PS: Clinical trial patient should call trial nurse first.
→ Consult subspecialist (ex: hepatologist, pulmonologist, endocrinologist) who is well known the irAEs
→ In refractory case, consider tumor necrosis factor alpha (TNFa) antagonists, mycophenolate or tacrolimus.
→ Long-term (>6 weeks) treatment with immunosuppressive drugs or use of infliximab increases the chance of opportunistic infections, such as Pneumocystis.

- High risk group for severe irAEs
  → Organ transplant patient
  → Pre-existing autoimmune disease
  → Prior episode of irAEs
  → Combination use of immune checkpoint inhibitors
- Exams and blood tests

	Baseline	***Every cycle***	***Q3M at least***
Routine	CBC/DC BUN/Cr. Alk-P AST/ALT/T-Bil/r-GT Sugar/HbA1C TSH/ Free T4/Cortisol Na/K/Ca/P/Mg LDH HBsAg/Anti-HBs/Anti-HBc/Anti-HCV EKG/CXR Whole body CT(&brain)	CBC/DC BUN/Cr. AST/ALT/T-Bil Sugar TSH Na/K/Ca/P/Mg	CBC/DC BUN/Cr. Alk-P AST/ALT/T-Bil/r-GT Sugar/HbA1C TSH/ Free T4/Cortisol Na/K/Ca/P/Mg LDH EKG/CXR Whole body CT (including brain)
Additional	Amylase/lipase T3/ACTH GH/FSH/LH Testosterone/PRL Anti-TPO Ab/ANA/IgG4	Alk-P/ r-GT Free T4/Cortisol LDH EKG/CXR	Amylase/lipase T3/ACTH GH/FSH/LH c-Peptide Testosterone/PRL Anti-TPO Ab/ANA/IgG4
Considered	PET/MRI of Brain/MRI of Sella turcica		PET/MRI of Brain/MRI of Sella turcica

玖、G-CSF 使用 Guideline

2006 Update of Recommendations for the Use of White Blood Cell Growth Factors: An Evidence-Based Clinical Practice Guideline **(Reference: *J Clin Oncol 2006, 24:1-19)***

Summary of 2006 Recommendations for the Use of CSF

Setting/Indication	Recommendation
Primary prophylaxis	1. High risk of FN based on age, medical history, disease characteristics, and myelotoxicity of the chemotherapy regimen. 2. For "dose-dense" regimens (Clinical trial data support the use of CSF when the risk of FN is in the range of 20% or higher)
Primary prophylaxis: Special circumstances	Age > 65 years; poor performance status; previous episodes of FN; extensive prior treatment including large radiation ports; administration of combined chemoradiotherapy; bone marrow involvement by tumor-producing cytopenia; poor nutritional status; the presence of open wounds or active infections; more advanced cancer, as well as other serious comorbidities (may cause prolonged neutropenia)
Secondary prophylaxis	patients who experienced a neutropenic complication from a prior cycle of chemotherapy (for which primary prophylaxis was not received), in which a reduced dose may compromise disease-free or overall survival or treatment outcome. (In many clinical situations, dose reduction or delay may be a reasonable alternative.)
Therapeutic use: Afebrile neutropenia	*not* routinely used
Therapeutic use: Febrile neutropenia	*High-risk: prolonged (>10 days) and profound (<0.1 x 10^9/L) neutropenia, age > 65 y/o, uncontrolled primary disease, pneumonia, hypotension and multi-organ dysfunction (sepsis syndrome), invasive fungal infection, or being hospitalized at the time of the development of fever.*
Dose intensity/density of chemotherapy	Dose-dense regimens should only be used within an appropriately designed clinical trial or if supported by convincing efficacy data.

Abbreviations: CSF, colony-stimulating factors; FN, febrile neutropenia; PBPC, peripheral-blood progenitor cell; AML, acute myeloid leukemia; MDS, myelodysplastic syndrome; ALL, acute lymphocytic leukemia; CHOP, cyclophosphamide, doxorubicin, vincristine, prednisone; G-CSF, filgrastim; GM-CSF, sargramostim; pegylated G-CSF, pegfilgrastim.

Summary of 2006 Recommendations for the Use of CSF (continued)

Setting/Indication	Recommendation
Adjuncts to progenitor-cell transplantation	*mobilize PBPC* often in conjunction with chemotherapy →Administration after autologous, but not allogeneic, PBPC transplantation is the current standard of care.
AML: Initial or repeat Induction chemotherapy	CSF use following initial induction therapy is reasonable →Although no favorable impact on remission rate, remission duration, or survival →Patients > 55 years of age may be most likely to benefit from CSF use
AML: CSF for priming effects	Use of CSF for priming effects is *not* recommended.
AML: Consolidation chemotherapy	recommended after the completion of consolidation chemotherapy because of the potential to decrease the incidence of infection and eliminate the likelihood of hospitalization in some patients receiving intensive post-remission chemotherapy →More profound shortening of the duration of neutropenia after consolidation chemotherapy for patients with AML in remission than for patients receiving initial induction therapy →As yet there is no information about the effect of longer-acting pegylated CSFs in patients with myeloid leukemias, and they should *not* be used in such patients outside of clinical trials.
MDS	Intermittent administration of CSF may be considered in a subset of patients with severe neutropenia and recurrent infection.
ALL	CSF administration is recommended after the completion of the initial first few days of chemotherapy of the initial induction or first post-remission course, thus shortening the duration of neutropenia of < 1,000/mm^3 by approximately 1 week.
Acute leukemia in relapse	CSF should be used judiciously, or not at all, in patients with refractory or relapsed myeloid leukemia since the expected benefit is only a few days of shortened neutropenia.

Radiotherapy ± chemotherapy	CSF should be *avoided* in patients receiving concomitant chemotherapy and radiation therapy, particularly involving the *mediastinum.* →In the absence of chemotherapy, therapeutic use of CSF may be considered in patients receiving radiation therapy alone if prolonged delays secondary to neutropenia are expected.
Older patients	Prophylactic CSF for patients *aged ≧ 65 years* with lymphoma treated with curative chemotherapy (CHOP or more aggressive regimens) should be given to reduce the incidence of FN and infections.
Comparative clinical activity of G-CSF and GM-CSF	No guideline recommendation can be made regarding the equivalency of the two colony-stimulating agents. →Further trials are recommended to study the comparative clinical activity, toxicity, and cost-effectiveness of GCSF and GM-CSF.
Treatment for radiation injury	Current recommendations for the management of patients exposed to lethal doses of total-body radiotherapy, but not doses high enough to lead to certain death due to injury to other organs

◎ **健保給付規定:**

白血球生長激素(G-CSF)(101/6/1 updated)：

§ 短效型注射劑（如 filgrastim、lenograstim）：

1.限

(1)用於造血幹細胞移植患者。

(2)血液惡性疾病接受靜注化學治療後。

(3)先天性或循環性中性白血球低下症者(當白血球數量少於 1000/cumm，或中性白血球 (ANC) 少於 500/cumm)。

(4)其他惡性疾病患者在接受化學治療後，曾經發生白血球少於 1000/cumm，或中性白血球 (ANC) 少於 500/cumm 者，即可使用。

(5)重度再生不良性貧血病人嚴重感染時使用，惟不得作為此類病人之預防性使用 (86/9/1)。

(6)*化學治療，併中性白血球缺乏之發燒，若中性白血球小於 100/cumm、癌症不受控制、肺炎、低血壓、多器官衰竭或侵犯性黴菌感染等危機程度高之感染。*

(7)對於骨髓造血功能不良症候群（MDS）的病人，若因嚴重性的中性白血球過低（ANC<500/cumm）而併發感染時，可間歇性使用 G-CSF，但不得作為長期且常規性使用。

(8)週邊血液幹細胞的趨動─不論在自體或異體幹細胞的收集，應於收集前之 4~5 日開始皮下注射 G-CSF，其劑量為 10μg /KG/day。

2.患者如白血球超過 4000/cumm，或中性白血球超過 2000/cumm 時，應即停藥。惟當預估其骨髓功能不易恢復時，雖其血球已達上述標準，仍可給予半量之治療，若仍可維持血球數，則可給予四分之一劑量，若仍可維持血球數，則停用。任何時候，若白血球或中性白血球數過度增高，即應停藥。

§ 長效型注射劑(如 pegfilgrastim)：

限非骨髓性癌症合併有骨髓侵犯之患者，在骨髓抑制性抗癌藥物治療後，且曾經發生白血球少於 1000/cumm，或中性白血球 (ANC) 少於 500/cumm 者使用。

壹拾、 Definition of objective response in solid tumors according to RECIST criteria (According to RECIST criteria 1.1)

- **Unidimensional evaluation**
- **Measuring tumor burden**

Clarify minimum size measurable non-nodal lesions

Up to FIVE measurable lesions (TWO per organ)

Select largest most reproducible target lesion

- **Nodal evaluation**

  Measure SHORT axis•

  Add ACTUAL short axis measurement to sum of longest diameters of non-nodal lesions

- **Evaluation of target Lesion**
- **Complete Response (CR)**

  Disappearance of all target lesions, any pathological lymph nodes (whether target or non-target) must have reduction in short axis to <10 mm.

- **Partial Response (PR)**

  At least a 30% decrease in the sum of diameters of target lesions, taking as reference the baseline sum diameters

- **Progressive Disease (PD)**

  At least a 20% increase in the sum of diameters of target lesions, taking as reference

  The smallest sum on study (this includes the baseline sum if that is the smallest on study). In addition to the relative increase of 20%, the sum must also demonstrate an absolute increase of at least 5 mm. (Note: the appearance of one or more new lesions is also considered progression).

- **Stable Disease (SD)**

  Neither sufficient shrinkage to qualify for PR nor sufficient increase to qualify for PD, taking as reference the smallest sum diameters while on study

- **Evaluation of non- target Lesion**
- **Complete Response (CR)**

  Disappearance of all non-target lesions and normalization of tumor marker level. All lymph nodes must be non-pathological in size (<10 mm short axis).

- **Non-CR/Non-PD**

  Persistence of one or more non-target lesion(s) and/or maintenance of tumor marker level above the normal limits.

- **Progressive Disease (PD)**

  Unequivocal progression of existing non-target lesions. (Note: the appearance of one or more new lesions is also considered progression).

Table 1 – Time point response: patients with target (+/– non-target) disease.

Target lesions	Non-target lesions	New lesions	Overall response
CR	CR	No	CR
CR	Non-CR/non-PD	No	PR
CR	Not evaluated	No	PR
PR	Non-PD or not all evaluated	No	PR
SD	Non-PD or not all evaluated	No	SD
Not all evaluated	Non-PD	No	NE
PD	Any	Yes or No	PD
Any	PD	Yes or No	PD
Any	Any	Yes	PD

CR = complete response, PR = partial response, SD = stable disease, PD = progressive disease, and NE = inevaluable.

Table 2 – Time point response: patients with non-target disease only.

Non-target lesions	New lesions	Overall response
CR	No	CR
Non-CR/non-PD	No	Non-CR/non-PD[a]
Not all evaluated	No	NE
Unequivocal PD	Yes or No	PD
Any	Yes	PD

CR = complete response, PD = progressive disease, and NE = inevaluable.

a 'Non-CR/non-PD' is preferred over 'stable disease' for non-target disease since SD is increasingly used as endpoint for assessment of efficacy in some trials so to assign this category when no lesions can be measured is not advised.

拾、化學治療 Guideline

化學治療藥物開立

1. 化學治療的 therapeutic index 很窄。潛在的化學治療的併發症從一般輕微反應到死亡皆有可能。因此，與其他藥物處方相比較，化學治療的給予需要更多的事前準備與評估。在開立化學治療藥物之前，建議：
   (1). 確診為癌症病患
   (2). 選用化學處方合適於此診斷，常見處方可由「癌症中心資源查詢網」常用之 Adjuvant Chemotherapy Regimens 查詢資料(http://lnkwww.cgmh.org.tw/intr/intr2/c33j00/adjuvant/adjuvant.htm)
   (3). 病患健康狀況得足以經得起化學治療的副作用

甲、 估計的生存期限大於 3 個月

乙、 ECOG 0，1，或者 2

丙、 足夠的檢驗值(WBC > 3000 [ANC>1500]), 血小板>100k。 AST，ALT，和< 3 x upper limit。 Bilirubin <1.5 x。 Creatinine <2 mg/dl)

丁、 目前無未被控制的感染

戊、 個別狀況須與主治醫師討論。

2. 向病人及家屬說明治療的目的、時間、流程、藥物作用、副作用等注意事項，並取得同意及簽署化學治療同意書
3. 化學治療劑量計算以身體表面積(body surface area, BSA)為單位。如果病患的身體表面積超過 2，化學治療劑量通常局限於 2 的 BSA
4. 化學治療電腦醫囑開立流程參考癌症中心資源網
5. 化學治療醫囑開立之前需與主治醫師確定處方與劑量並以電腦醫囑開立
6. 電腦醫囑開立後請主治醫師或專科護理師覆查後才送單

化學治療藥物注射之執行

1. 核對醫囑，確立已簽署化學治療同意書
2. 執行小量靜脈注射化學藥物時須戴 N95 口罩、醫用檢查手套、防水不織布藥品袖調配衣及護目鏡
3. 執行病人辨識

   病人辨識方法有以下幾種，請病人或家屬說出其名字、核對病人健保卡或身分證、確認其床頭牌或手圈，以**雙重**確認途徑來執行病人辨識。
4. 測試回血以確定靜脈留置針或 PORT-A 彎針在血管內，且功能正常。

   起皰性化學藥物外漏時易造成皮膚、血管、組織纖維化及壞死；刺激性化學藥物外漏時易造成注射處周圍血管發紅、緊繃感、疼痛不適等，故注射下列藥物時，必須先確定靜脈留置針在血管內且其功能正常
5. 每給予 1~2 cc 化學藥物後，需再反抽測試回血，以確定針頭位置是否正確

   從週邊靜脈給予起皰性化學藥物時，需觀察周圍皮膚是否腫脹以確定藥物是否輸入病人血管內
6. 注射化學藥物時需隨時詢問病人是否有灼熱感或疼痛或腫脹等不適反應
7. 觀察病人注射過程及注射後的反應，如有副作用及過敏反應，應立即停止給藥並測量生命徵象
8. 凡接觸過化學藥物之廢棄物處理方式：可燃性如化學治療用物之針頭、空針、N95 口罩、醫用檢查手套、防水不織布藥品長袖調配衣、紗布及護目鏡等可燃性廢棄物，與不可燃物品，如：化學藥物點滴瓶(IV set 不需另外拔除)，必需先以高密度 PE 袋封好，棄置於基因毒性廢棄物垃圾桶內（套有紅色感染性 PE 袋之防漏、防刺與加蓋的容器(紙箱)內）；化學治療廢棄物需包裝二層

拾壹、癌症化學治療藥物外漏的處置(Extravasation)

一、所謂外漏現象主要指一般靜脈輸注液或藥物輸注時，因針頭位置脫離至靜脈血管外，以致使藥物外漏至周圍皮下組織或自血管滲透外溢，進而導致患部肌肉組織的傷害或壞死，此時病人的注射部位會感覺到刺痛、燒灼感，外觀上可見腫脹,起水泡及發紅等現象

- 化學治療藥物外漏所引起的局部毒性，可以分為起泡性和刺激性兩大類：
  - *起泡性(Vesicant)化學治療藥物*

    是指化學治療藥物外漏時，會使組織產生水泡而逐漸破壞皮膚組織，並產生肌肉組織壞死的現象

成分名	商品名
Dactinomycin	Cosmegen/Actinomycin-D
Daunorubicin	Cerubidine/Daunomycin
Doxorubicin	Adriamycin
Epirubicin	Pharmorubicin
Idarubicin	Idamycin/Zavedos
Mechlorethamine	Mustargen
Mitomycin	Mutamycin
Vinblastine	Velban
Vincristine	Oncovin
Vindesine	Eldisine

- 化學治療藥物外漏若在注射部位發生外漏，則會引起強烈的發炎反應，約在幾個小時內會明顯出現紅腫、疼痛，在接下來的幾個星期可能演變成深層皮膚的潰爛及局部組織壞死
  →起泡性(Vesicant)化學治療藥物注射時必須非常小心，因此類藥物外漏時所造成的傷害是最嚴重的！
  - *刺激性(Irritant)化學治療藥物*
    是指當注射或滴注此種藥物時，會引起注射部位的血管周圍感到疼痛或沿著整個血管都有疼痛的感覺，但卻無皮膚組織壞死的現象

成分名	商品名
Bleomycin	Blenoxane/Bleocin
Carmustine	BCNU
Cisplatin	Platinol/Platamine
Dacarbazine	DTIC-Dome
Etoposide	VePesid/VP-16
Fluorouracil	Adrucil/5-FU
Menogaril	Tomosar
Mitoxantrone	Novantrone
Paclitaxel	Taxol
Streptozocin	Zanosar
Teniposide	Vumon

- 有些學者認為 paclitaxel 只具局部刺激性，但近年有些臨床報告卻認為是屬於起泡性的，因為 paclitaxel 在外漏的初期 24 小時內並無任何嚴重的症狀，但在第 11 天後病人在注射部位有紅腫和壞死的現象，並伴有嚴重局部周邊神經病變(peripheral neuropathy) ，而必須施行清創(debridement)和修補(closure)的外科重整手術，但由於無大型的臨床經驗分析報告，故仍有待進一步的評估

二、一般而言，化學治療藥物可能引起外滲的評估反應可分為：

- *靜脈刺激(irritation of the vein)*
  - 注射的部位引起疼痛不適和緊繃感，且沿著注射部位的血管可能會引起發炎反應，但它不會引起組織的潰瘍
  - 通常此現象在藥物被稀釋，濃度降低或緩慢滴注,即可避免或減輕
- *熱反應(flare reaction)*
  - 為一種局部的過敏反應，可能和 anthracycline 抗生素化學治療藥物或 nitrogen mustard 有關，

可在注射後 30 到 60 分鐘內發生局部血管的搔癢及紅腫，是皮膚反應現象中最輕微的，且不會造成長期的傷害

- 熱反應多發生在血管狀況不佳或血管過細的病人,及先注射類固醇藥物後再注射 anthrcycline 類或 nitrogen mustard 的化學治療藥物，通常引起的局部血管紅腫、疼痛、搔癢，在注射 30 分鐘後症狀會逐漸消失

三、

外滲藥物	冷/熱敷	解毒劑/劑量
Amsacrine	冷敷	1.Sodium bicarbonate(8.4%) 5ml SC + Dexamethasone(4mg/ml) 1ml SC 2.DMSO(50-90%) 15ml q2-4h for 3 days
Carmustine	冷敷	Sodium bicarbonate(8.4%) 5ml SC
Cisplatinum	冷敷	Sodium thiosulfate(10%) 4ml SC + sterile water for injection 6ml
Dactinomycin	冷敷	1.Sodium thiosulfate(10%) 4ml SC or Ascorbide acid (250mg/ml) 0.2ml (make up to 1ml) 2.DMSO(50-90%) 15ml q2-4h for 3 days
Daunorubicin	冷敷	1.Sodium bicarbonate(8.4%) 5ml SC + Dexamethasone(4mg/ml) 1ml SC 2.DMSO(50-90%) 15ml q2-4h for 3 days
Doxorubicin	冷敷	1.Sodium bicarbonate(8.4%) 5ml SC + Dexamethasone(4mg/ml) 1ml SC 2.DMSO(50-90%) 15ml q2-4h for 3 days
Idarubicin	冷敷	1.Sodium bicarbonate(8.4%) 5ml SC + Dexamethasone(4mg/ml) 1ml SC 2.DMSO(50-90%) 15ml q2-4h for 3 days
Epirubicin	冷敷	1.Sodium bicarbonate(8.4%) 5ml SC + Dexamethasone(4mg/ml) 1ml SC 2.DMSO(50-90%) 15ml q2-4h for 3 days
Etoposide	熱敷	1.Sodium bicarbonate(8.4%) 5ml SC 2.Hyaluronidase(150IU/ml) 3-5ml SC
Mechlorethamine (Nitrogen mustard)	冷敷	Sodium thiosulfate(10%) 4ml SC + sterile water for injection 6ml
Mitomycin	冷敷	1.Sodium thiosulfate(10%) 4ml SC or Ascorbide acid (250mg/ml) 0.2ml (make up to 1ml) 2.DMSO(100%) 1 time 3.Pyridoxine(100mg/ml) SC
Vinblastine	熱敷	1.Sodium bicarbonate(8.4%) 5ml SC 2.Hyaluronidase(150IU/ml) 3-5ml SC
Vincristine	熱敷	同上
Vindesine	熱敷	同上
Vinorelbine	熱敷	同上

DMSO=Dimethyl sulfoxide

- 當病人主訴注射化學治療藥物時有疼痛、燒灼感或血管不見回血時，即使發現病人肌肉組織沒有腫脹的現象，仍應考慮是否有外滲的情況發生
- 若不幸發生外滲，其處理方式可依外滲藥物種類有否解毒劑而不同；若沒有解毒劑或無法反

抽外漏的液體，則將針頭拔出；若有解毒劑，則依上表處理

- 其中熱敷為適用於 alkaloid 類的藥物，其需每 30-60 分鐘施予一次熱敷再休息 15 分鐘，如此重複共一天
- 冰敷則適用於其他藥物，在外滲的部位給予冰敷一小時，每天共四次,需連續冰敷 3-5 天

四、謹慎選擇給藥的部位

- 由末梢血管來給藥以前臂之大血管為佳，如後基底靜脈，其次以手臂的血管為主，因其較容易注射及固定，而腕關節和肘前窩區則富含神經及肌腱，應避免由此處給藥
- 若已知病人需接受多次的化療，應建議裝置人工靜脈血管(Hickman、Goshong 導管、Mediport、Port-A catheter 等)
- 為保持血管的完整性，注射部位移由遠端開始注射，同時血管必須輪流交替使用

五、

處理人員	處理步驟
施打化學治療藥物之人員 (醫師、護士)	1. 檢查化學治療藥物是否外滲及觀察症狀 •注射部位有疼痛及燒灼感 •針頭推插入部位或沿著靜脈出現種 •外滲部位出現紅腫 •靜脈注射部位流速減緩 •針頭刺入後未見回血 2. 停止靜脈注射但勿拔出針頭套管(cannula) 3. 將針頭與注射管線分離,並接裝 10ml 針筒，勿拔出針頭 4. 由皮下組織盡量抽取藥物，血液及組織液至少 3-5ml 5. 注射 10-15ml N/S 至外滲處 6. 若外滲之藥物沒有解毒劑，則將針頭拔出;若有解毒劑，則固定針頭準備解毒劑
化學治療專科醫師、臨床藥師	1. 用小號針頭及針筒(如 insulin 用 25 號針)吸取皮下小泡中的藥物及組織液 2. 選用適當的解毒劑 3. 由原已存在的針頭或靜脈點滴導管注射解毒劑後,再注入 decadron 或 solucortef 以減少發炎現象 4. 注射 decadron 或 solucortef 時,應依照東南西北反時鐘的方向依序每次注射類固醇藥物總劑量的 1/4 量，以保護藥品能夠迅速的穿透到外滲部位以利藥物的吸收 5. 施打解毒劑前確定針頭是否位於皮下位置而不是在靜脈中，並在注射完後即可將針頭拔出，但勿對患部壓迫 6. 紅腫部位可施予類固醇軟膏一天兩次，但若有感染之慮時,可改用含抗生素之複方軟膏，但不可包紮傷口部位 7. 紀錄病人整個外滲狀況及處理情形，並完成特殊異常意外事件報告 8. 必要時可拍攝外滲部位，以比較外滲後組織傷害的反應過程
照護化學治療藥物外滲病人的醫護人員	1. 定時更換消毒紗布 2. 如外滲部位發生在四肢，可以枕頭墊高至腫脹消失為止 3. Alkaloid 類的化學治療藥物需施予每 60 分鐘一次熱敷 4. 除 alkaloid 類的化學治療藥物外，在外滲部位給予冰敷一天四次，每次一小

	時，共 3-5 天 5. 固定回診，以觀察患部組織的變化，並記錄復原或惡化的情況 6. 如果局部組織持續惡化或疼痛未能緩解，應會診外科或整形外科專科醫師，評估是否需給予外科清創及修補

拾貳、Catheter related infection 管路相關感染

Figure1. Methods for the diagnosis of acute fever for a patient suspected of having short-term central venous catheter infection or arterial catheter infection. CFU, colony-forming units

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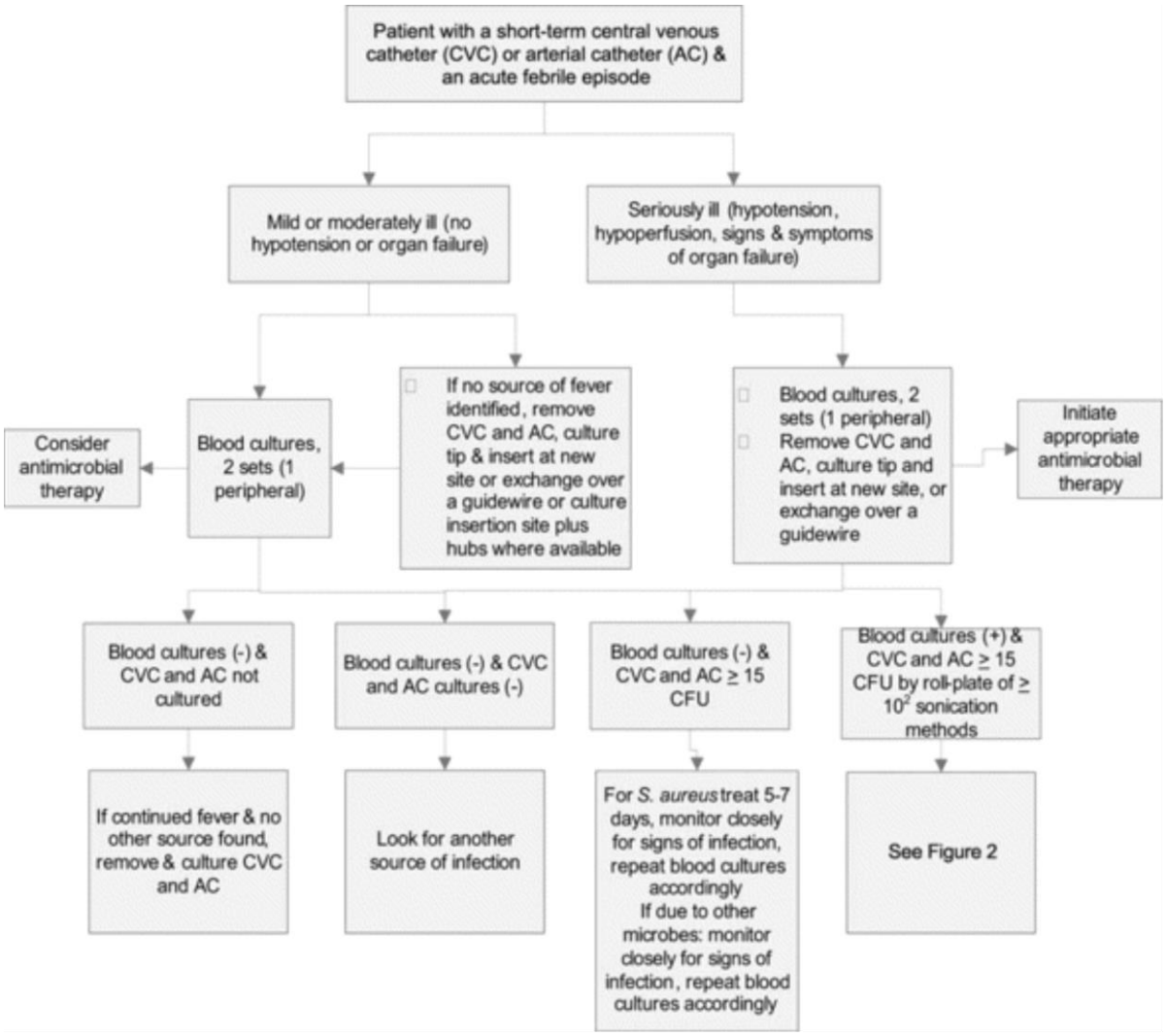


Figure2. Approach to the management of patients with short-term central venous catheter-related or arterial catheter-related bloodstream infection. CFU, colony-forming units; S. aureus, Staphylococcus aureus.

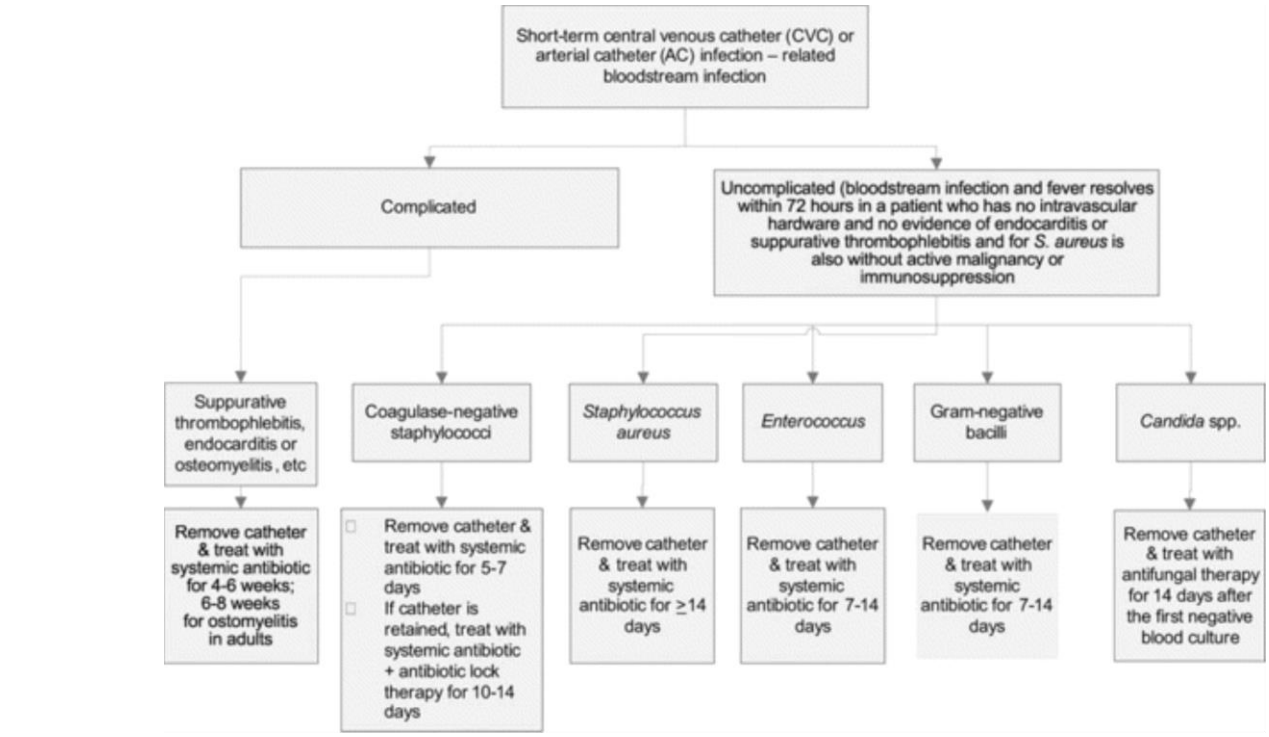
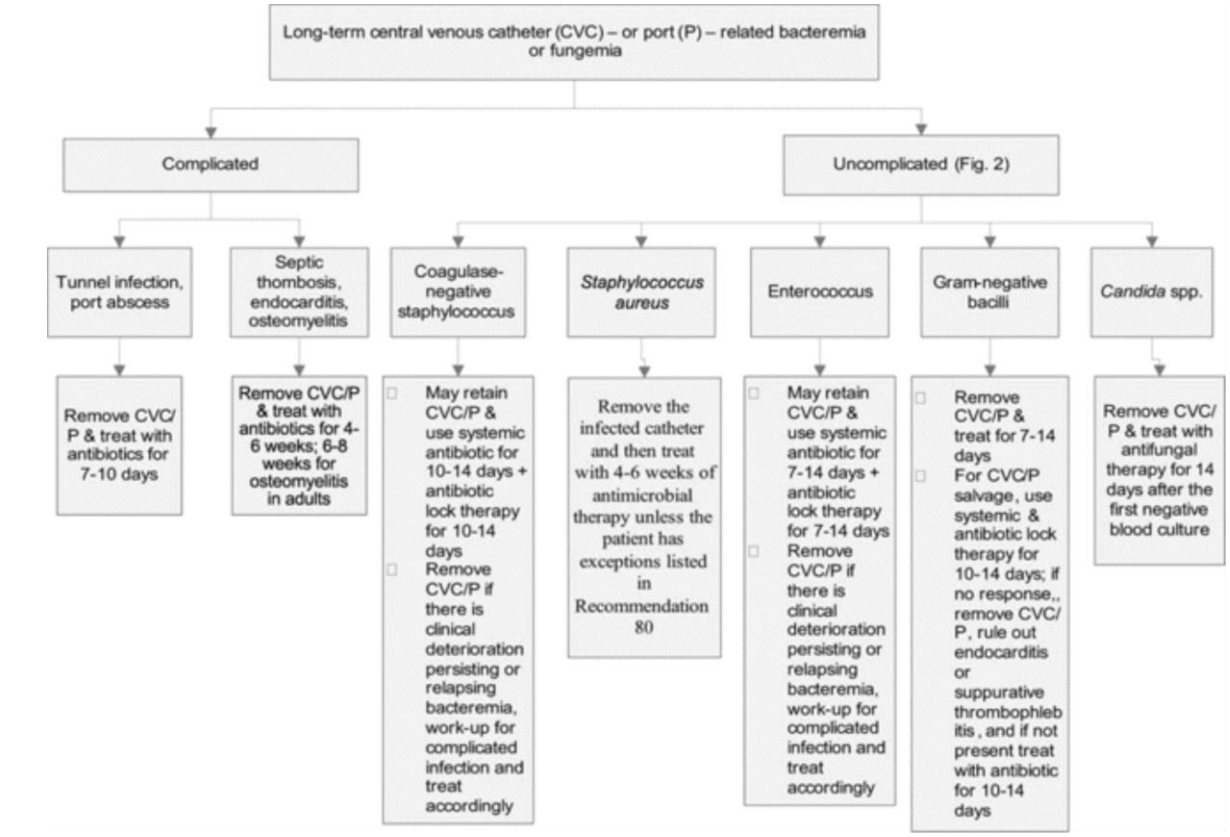


Figure3. Approach to the treatment of a patient with a long-term central venous catheter (CVC) or a port (P)-related bloodstream infection.



Reference: Clinical Practice Guidelines for the Diagnosis and Management of Intravascular Catheter-Related Infection: 2009 Update by the Infectious Diseases Society of America

拾參、Oncology emergencies 腫瘤急症處理

➢ Oncologic Emergencies

1. Structive－Obstructive emergencies
2. Metabolic emergencies
3. Treatment-related emergencies

➢ Structural Obstructive Oncologic Emergencies

1. Superior Vena Cava Syndrome
2. Cardiac Tamponade
3. Intestinal Obstruction
4. Urinary Obstruction
5. Malignant Biliary Obstruction
6. Spinal Cord Compression
7. Increased Intracranial Pressure
8. Upper Airway Obstruction

➢ Superior Vena Cava Syndrome

1. Life threatening: upper airway obstruction
2. Pathophysiology: tumor invasion, external compression, or thrombosis of SVC
3. Malignancy associated (most common): lung cancer: 75%, lymphoma: 15%
4. Diagnosis: clinical symptoms and sign plus mediastinal mass
5. Superior vena caca syndrome symptoms and signs

Symptoms %		Signs %	
Dyspnea	63%	Venous dilation of neck	66%
Facial swelling or head fullness	50%	Venous dilation of chest wall	54%
Cough	24%	Facial edema	46%
Arm swelling	18%	Cyanosis	20%
Chest pain	15%	Plethora of face	19%
Dyaphagia	9%	Edema of arm	14%

6. Superior Vena Cava Syndrome - Treatment
   a) Generalized: diuretics with low salt diet, head elevation, oxygen supplement
   b) Radiotherapy: non-small lung cancer and metastatic cancer
   c) Chemotherapy: small cell lung cancer and lymphoma
   d) Surgery: bypass grafts and radical excision with vein grafting

➢ Cardiac Tamponade

1. Pericardial effusion in malignancy – most common: lung and breast cancer
   Cardiac tamponade:
2. Limited diastolic filling of ventricle – result in low cardiac output
3. Clinical S/S – Depend on
   a) Rate of fluid accumulation
   b) Volume of the fluid

c) Underlying cardiac function

4. Cardiac Tamponade - Diagnosis
   a) Symptoms: dyspnea, orthopnea, cough, chest pain, weakness
   b) Signs: pleural effusion, jugular venous engorement, hepatomegaly
   c) Physical examinations: paradoxical pulse, Kussmaul’ s sign, distant heart sound
   d) CXR: cardiac enlargement – water bottle figure
   e) EKG: low ORS voltage, electric alternas
   f) Echocardiogram: most effective, simple diagnostic method
      # R’ t atrial and ventricular collapse
5. Cardiac Tamponade - Treatment
   a) Generalized: oxygen, fluid supply, monitor blood pressure
   b) Pericardiocentesis
   c) Subxyphoid drainage
   d) Pericardial-pleural window
   e) Percutaneous tube, pericardiostomy+ pericardial sclerotherapy
   f) Percutaneous balloon tube pericardiostomy

➢ Intestinal Obstruction

1. iagnosis: standing plain abdomen – multiple air-fluid level or dilatation of small and large bowel
2. Treatment:
   a) Fluid and electrolyte support
   b) Decompression
   c) If obstruction not improved – bypass surgery or colostomy
   d) If condition stablized – further definite therapy

➢ Obstructive Uropathy

1. Bilateral ureteral obstruction – oliguria, anuria -> uremia
2. Causes: Malignant tumor: cervix, rectum, prostate, bladder or metastatic cancer
   Sequela of anti-tumor therapy: surgery, post-radiotherapy fibrosis
3. Obstructive Uropathy - Diagnosis
   a) Renal echo: quickly and effective
   b) CT scan: obstructive level, tumor extent
   c) Cystoscope, retrograde pyelography: safe in azotemia patients
   d) Percutaneous nephrostomy with antegrade pyelography: level and nature of obstruction, drainage
4. Obstructive Uropathy - Treatment
   e) Percutaneous nephrostomy tube
   f) Ureteral stent
   g) Ureteral diversion
   h) Supra-pubic cystostomy
   i) Emergent hemodialysis

➢ Malignant Biliary Obstruction

1. Symptoms: jaundice, light-colored stool, dark urine, pruritus, weight loss
2. Causes: Primary tumor: pancreas, ampulla vater, bile duct, liver tumor
   Metastatic tumor: stomach, colon, breast, lung cancer

3. Diagnosis:
   a) Abdominal sonography
   b) Abdominal CT scan
   c) Percutaneous transhepatic cholangiography (PTC)
4. Treatment:
   a) PTCD
   b) Biliary stenting
   c) Surgery bypass
   d) Radiotherapy with or without chemotherapy

➢ Upper Airway Obstruction

1. Causes: Head and neck cancer – larynx, pharynx; Primary or metastatic lung cancer Lymphoma; Primary trachea cancer; Mediastinal tumor
2. Symptoms and Signs: dyspnea, stridor, hoarseness inspiratory stridor
3. Diagnosis: ENT consultation; Radiologic evaluation: locate and define airway obstruction
4. Treatment: Emergent tracheostomy; humidification; airway care; Steroid Antibiotics; Further definite therapy

➢ Spinal Cord Compression

1. Irreversible neurological damage unless emergent management
2. 95% extramedullary
3. Most common primary sites: lung, breast
4. Site distribution:
   Throacic: 70%
   Lumbosarcral: 20%
   Cervical: 10%
   Style: most common from vertebral body or pedicle, intervertebral foramen
5. Frequency of Tumor Types With Cord Compression :
   breast cancer 20.6 %; lung 17 %, lymphoma 9 %, prostate 7%, sarcoma 6.6 %, myeloma 6 % kidney 5.6 % other 27 %
6. Spinal Cord Compression – Symptoms :
   Pain: early symptoms, local or radical
   Motor weakness: ambulatory, paraparesis, paraplegia
   Sensory loss: numbness, tingling, loss of sensory
   Autonomic dysfunction: bowel and bladder dysfunction
7. Spinal Cord Compression - Signs
   Vertebral tenderness, Abnormal below the level of cord compression
   Deep tendon reflex, Babinski sign, Motor weakness and atrophy
   Hypersthesia, Decrease anal sphincter, Cauda equina syndrome
   Ataxia: very rare
8. Spinal Cord Compression – Pathogenesis
   By extension from vertebrae (85%)
   By extension from para-vertebral tumor (10%)
   Occlusion of spinal arteries (possible cause for acute deterioration)
   Hematogenous: from Baston’s venous plexus
9. Spinal Cord Compression - Diagnosis

Plain X ray, Myelography, CT scan, MRI, Bone scan

10. Spinal Cord Compression – Surgical Indication
    Histology unknown
    Relapse after radiotherapy
    Neurological deterioration during radiotherapy
    Radioresistant tumor
    Pathologic fracture with spinal instability
11. Spinal Cord Compression – Radiotherapy
    Hypofractioated radiotherapy
    Portal: above or below 2 vertebral body. A single posterior field
12. Spinal Cord Compression – Steroids
    Anti-edema effect
    Dose: not established, but favor Initial high dose dexamethasone
13. Spinal Cord Compression – Prognostic Factors
    a) Motor status: Ambulatory: better; Paraplegia: poor
    b) Onset of neurological dysfunction: Rapid: poor; Slow: better
    c) Loss of sphincter control: poor
    d) Location: poor in proximal thoracic spine
    e) Tumor type: Radioresistant: poor; Radiosensitive: better

➢ Increased Intra-cranial Pressure

1. Causes: Primary brain tumors
   Metastatic brain tumors
   Meningeal carcinomatosis
   Metastatic cancers: lung and breast cancer – most common
2. Symptoms and signs: headache, mental status change, weakness, focal neurological deficit, papilledema
3. Diagnosis: CT scan or MRI
   Lumbar puncture: if meningeal carcinomatosis is suspected
4. Treatment
   Dexamethasone, Glycerol or mannitol, Radiotherapy, Craniotomy
   Intrathecal chemotherapy: for meningeal seeding
   Others: anti-convulsant for seizure, hyperventilation for coma patients

➢ Metabolic Emergencies

1. Metabolic emergencies including : Hypercalcemia, SIADH
   Lactic acidosis, Hypoglycemia, Adrenal insufficiency
2. Hypercalcemia in Malignancy
   Calcium source: mainly via bone
   Common associated malignancy: lung, breast, myeloma, head and neck, kidney, esophagus
3. Mechinism of hypercalcemia:
   a) Osteoclastic bone resportion more prominent
   b) 1,25(OH)2 VitD usually suppressed
   c) Alkalotic rather than hyperchloremic and mild acidotic
   d) Not elevated iPTH
   e) Bone resorptive factors: PTH-related protein, cytokines, growth factor

4. Symptoms and Signs of hypercalcemia
   General: weakness, lethargy
   Gastrointestinal: anorexia, nausea, vomiting, constipation
   Renal: polydipsia, polyuria, renal failure
   Cardiac: arrhythmia(EKG: short QT interval, broad T wave)
   Neurologic: hyporeflexia, stupor, coma
5. Hypercalcemia in Malignancy
   Total calcium adjusted from serum albumin
   corrected calcium (mg/dL) = measured calcium + (4 – albumin (g/dL)) * 0.8
   Level of hypercalcemia
   mild: 10.5 – 12.5 mg/dL
   moderate: 12.5 – 14.5 mg/dL
   severe: >14.5 mg/dL
6. Treatment of hypercalcemia
   a) Intravenous fluids and diuretics
   b) Antiresorptive drugs: Bisphosphonates
   Amino-bisphosphonates (Pamidronate,Zoledronate) have greater antiresorptive Capabilities
   c) Cytotoxic therapy, e.g. R/T, C/T

➢ Treatment-related Emergency

Tumor Lysis Syndrome
Hemolytic-uremia Syndrome
Neuropenia and Fever
Pulmonary Infiltrates
Hemorrhagic Cystitis

拾肆、Pain control

➢ Three step "analgesics ladders by cancer unit of WHO

Mild-moderate cancer related pain – non-opioid analgesic +/- adjuvant analgesic
If failure - opioid +/- non-opioid or adjuvant analgesic (Codeine, Oxycodone, Propoxyphene)
Severe pain - opioid agonist +/- non-opioid or adjuvant analgesic

Categories of analgesic drugs

Nonopiate anagesics	Opiates	Adjuvant agents
Acetaminophe NSAID	Weak opiates Potent opiates Full agonist Partial agonist Agonist-antagonists	Heterocyclic antidepressants Anticonvulsant Psychotropic agents

* Systemically administered opioid therapy
- agonist, agonist-antagonist and antagonist pure agonist: codeine, morphine, methadone, meperidine
  # partial agonist: limited usefulness because of ceiling effect and withdrawal reaction

- Opioid selection

Pain of moderate or greater severity: Codeine, Oxycodone, Propoxyphene + Acetaminophen or NSAID

Severe pain: Morphine sulfate is usually preferred

Methadone: long half life

- not preferred for elderly or with major organ failure, difficult to titrate, delay toxicity

Meperidine (Demerol): not recommendation for management of cancer pain

- Because its half-life is short and its metabolite, normeperidine, is toxic
- Demethylated → Normepridine

Subtle adverse mood effect, tremulousness, multifocal clonus and occasionally seizure

➢ Tramadol:

- relatively new analgesic with a complex mode of action
- weak opioid activity at the mu opioid receptor, and it also active alpha2 receptors
- Analgesic activity→ its antidepressant-like ability to inhibit serotonin and norepinephrine reuptake at the synaptic level
- mild – moderate pain
- Relative potency of tramadol to morphine is about 1:5 to 1:4 for oral route

➢ Transdermal formulation, Fentanyl (Durogesic)

$ Minimally invasive, convenience

$ Drug reservoir separated from the skin by rate-limiting membrane to control drug delivery

$ Dose interval for each system usually 72hrs deliver 25,50,75,100 ug/hr

Durogesic (ug/h)	Daily fentany releasing dose (mg/day)	Eqivalent oral morphine dose (m/day)
25	0.6	60 (30-90)
50	1.2	120 (90-150)

$ Less nausea, vomiting, skin itching than morphine

$ Half-life: 16-24 hrs

$ 25 ug/h transdermal system is roughly eqianalgesic to iv morphine 1 mg/hr

➢ Equianalgesia: ratio of the dose of two analgesics required to produce the same analgesic effect

Reference tables: 10 mg of parenteral morphine

Potential of major opiates relative to morphine (Morphine = 1)

Codeine
Mepreidine
Methadone

Bioavailability of opiates when administered orally

Codeine
Meperidine
Methadone
Morphine

Half-lives of common use opiated

Codeine

Meperidine
Methadone
Morphine
Controlled-release morphine

➢ Selecting a route of system opioid administration

- Oral route is usually the most appropriate
  (convenient, cost-effective, and minimally invasive)
- Impaired swallowing or GI obstruction
  @ Alternative non-invasive route:: rectal, subligual, and transdermal routes
  # Parenteral route: IV, IM, SC

➢ Breakthrough pain

highly prevalent; supplemental rescue dose on an as-needed basis
rescue dose about 10-25% of 24-hour baseline dose, oral rescue doses offered up to every 1-2 hr, parenteral doses offered up to every 15-30 min
Total dose of as-needed rescue medication available in a specific interval should be equal to the regular dose given during that interval
eg. 90 mg of controlled-release morphine q12h should be given 30 mg of immediate-release morphine q4h for unrelieved, breakthrough pain
Frequent episodes (≥ 4 per day) suggest the need to increase regular dosing, which may prevent breakthrough pain

➢ Management of side effects

- Most common: constipation, nausea & vomiting, sedation; respiratory depression: rare
  Other dose-limiting adverse effects: sedation, delirium, myoclonus and respiratory depression

Constipation: softing agent, cathartic agent, osmotic laxative, colon lavage
Nausea & vomiting: antiemetics
Sedation: persist for days to weeks; reducing dose or use psychostimulant (caffeine or methylphenidate) or change to other drug or procedure
Cognitive impairment, delirum: response to haloperidol 0.5 to 1 mg
Myoclonic jerks: treated with clonazepam 0.25 to 0.5 mg orally tid
Respiratory depression: Naloxone 0.4 mg in 10 ml saline

➢ Principles of opioid use

1. Choose a well-tolerated opioid, based if possible on the analgesic history
2. Use morphine as the routine drug of choice for moderate/severe pain
3. Avoid opioids:
   a) that are difficult to use, eg. Methadone
   b) with excess side effects, eg. Codeine
   c) of poor analgesic efficacy, eg.meperidine
4. Use “ATC” (around the clock) dosing for continuous pain; prescribe “PRN” rescue dosing for intermittent (incident/breakthrough) pain
5. Identify indications for alternate routes of administration.
6. Quantify analgesic response in magnitude and duration and titrate dosing based on that

7. Review opioid doses regularly until pain is controlled
8. Individualize dosing, taking into account age, sex, and kidney, liver, and cognitive function
9. Identify dose-limiting side effects, and those at risk for respiratory failure
10. Anticipate constipation and neuropsychological side effects
11. Diagnose pain syndromes often associated with relative opioid resitance
12. Be aware of the equianalgesic doses of commonly use opioids both by drug and route of administration.

拾伍、腫瘤標誌的應用 Tumor Marker

1. 篩檢 (Screening)
2. 診斷 (Diagnosis)
3. 預後 (Prognosis)
4. 評估療效 (Monitoring therapy)
5. 偵測再發 (Detection of recurrence)

➢ 腫瘤標誌與相關癌症

- 乳癌: CA 15.3, CEA
- 肺癌: CEA
- 胃腸癌: CEA, CA 19.9
- 攝護腺癌: PSA
- 肝癌: AFP
- 卵巢癌: CA 125
- 生殖細胞癌: AFP, Beta-HCG
- 子宮頸癌: SCC

●CEA (胚胎抗原)

- 正常值 < 5 ng/ml
- 半衰期: 不定
- CEA 提高之情況

  惡性疾病：大腸直腸癌、胰臟癌、肺癌、胃癌、乳癌、卵巢癌、膽管癌

  非惡性疾病：抽菸、肺氣腫、大腸息肉、潰爛性大腸炎、胰臟炎、酒精性肝硬化

  非惡性疾病的 CEA 值常 < 10 ng/ml
- CEA 的應用
  1. 對篩檢或診斷並無幫助
  2. 術前 CEA 提高者 →易再發，預後差
  3. 評估療效
     a) 治療前 CEA 高者，預後差
     b) 經治療明顯下降者，預後好
  4. 偵測再發
- CEA 與各類癌症
  1. 分期 (stage)愈高者，CEA 值亦愈高
  2. 對大腸直腸癌，病理組織分化好者 CEA 較高
  3. 肝轉移者，CEA 更高
  4. CEA 高，也須將髓質性甲狀腺癌 (Medullary Thyroid Cancer) 當做鑑別診斷

●AFP (甲型胎兒蛋白)

- 正常值 < 20 ng/ml

- 半衰期: 4.5 天（3.5-6.0 天）
- 來源：1. 卵黃囊（Yolk sac）、2. 肝（liver）
- 肝炎流行區作為肝癌的篩檢，相當有用
- AFP 提高之情況
  1. 懷孕
  2. 肝癌
  3. 非精原性生殖細胞腫瘤（non-seminomatous germ cell tumor)
  4. 肝炎
  5. 胃癌或胰臟癌
  6. 其他癌症或肝轉移

●β-HCG

- 正常值 < 5 mIμ/ml
- 半衰期: 30 小時
- β-HCG 提高之情況
  1. 懷孕
  2. 滋養層細胞疾病 (Gestational trophoblastic disease)
  3. 精原性生殖細胞腫瘤（Seminomatous germ cell tumor)
  4. 非精原性生殖細胞腫瘤（Non-seminomatous germ cell tumor)
  5. 其他癌症

●CA-125

- 正常值 < 35 U/ml
  半衰期: 4-7 天
- 來源：
  1. 卵巢癌的表面蛋白
  2. 生殖上皮的衍生物 (輸卵管、子宮內膜、子宮頸)
  3. 間皮細胞 mesothelial cell (腹膜、肋膜、心包膜）
- CA-125 提高之情況
  1. 正常女性： 1%
  2. 良性卵巢疾病 : 6%
  3. 惡性疾病
     卵巢癌：80%　　乳癌: 12%　　大腸直腸癌: 21%　　胰臟癌：59%　　肺癌: 32%
  4. 間皮細胞反應
     結核性腹膜炎　　開腹手術後　　腹水　　肋膜積水

●CA-19.9

- 正常值 < 37 U/ml
- 提高之情況
  惡性疾病：胰臟癌、胃癌、大腸直腸癌
  非惡性疾病：良性胰臟疾病、肝病
  尤其胰臟癌、胃癌最佳，且比 CEA 好

●CA-15.3

- 正常值 < 30 U/ml
- 提高之情況
  乳癌

原發位：20%　轉移性：63%

非惡性疾病

良性乳房疾病：6%　　肝炎：30%

- 乳癌 CA-15.3 且比 CEA 好

●PSA（攝護腺特殊抗原）

- 正常值　< 4 ng/ml
- 半衰期:　2-3 天
- 提高之情況

  攝護腺癌　　攝護腺肥厚　　攝護腺發炎　　攝護腺癌做過處置後
- 攝護腺肥厚 PSA 值通常 < 10 ng/ml
- 篩檢（有些學者建議 50 歲以上男性，每年作 PSA 及肛診檢查各一次）
- Free/ Total PSA 在攝護腺癌的應用
  1. 攝護腺癌的 Free PSA 偏低
  2. Free/Total PSA %　　癌症的可能性

0-10%	56%
10-15%	28%
15-20%	20%
20-25%	16%
> 25%	8%

  3. PSA 邊緣性提高時, Free/Total PSA 可增加一些特殊度 (specificity)

拾陸、常見的 immunohistochemical staining

NSE (neuron specific endolase): neuroendocrine tumor
AE1/AE3: carcinoma
CK5/6 and p63: squamous carcinoma
Hepar-1: hepatocellular carcinoma
Alpha-inhibin: adrenocortical carcinoma
Vimentin: mensenchymal tumor
GFAP (gilial fibrillary acidic protein): glioma
Desmin: muscle
S-100: melanoma, Schwanoma
HMB-45: melanoma
Myoglobin, Actin, HHF-35: Rhabdomyosarcoma
Factor VIII-related antigen: Angiosarcoma
LCA (leucocyte common antigen): lymphoid tumor
Leu-Mi: Hodgkin's disease, some Tcell lymphoma
CD 10: Renal clear cell carcinoma
CD 30 (Ki-I): anaplastic large cell lymphoma
CD 20 and CD79A: B cell markers
CD 38 and CD 138: plasmacytic markers
CD 3: T cell markers
CD56: neuroendocrine carcinoma
CD 34: bone marrow stem cell
CD 117 (c-kit): GIST
CD 99: Ewing’s sarcoma and primitive neuroectodemal tumor

Basic immunohistochemcal work-up of carcinoma of unknown primary

Primary markers	
CK7-/CK20+	Colorectal and Merckel cell carcinoma
CK7+/CK20-	Lung, breast, thyroid, endometrial, cervical, and pancreatic carcinoma and cholangiocarcinoma
CK7+/CK20+	Urothelial, ovarian and pancreatic ca and cholangioca
CK7-/CK20-	Hepatocellular, renal cell, prostate, Squamous cell

Thyroid transcription factor-1 (TTF-1): thyroid cancer
lung adenocarcinoma, small-cell carcinoma
and neuroendocrine carcinoma

P63: Lung poorly differentiated squamous carcinoma
Urothelial carcinoma

Immunoperoxidase tumor-staining patterns useful in differential diagnosis of poorly-differentiated neoplasms

Tumor type	Immunoperoxidase staining
Carcinoma	Epithelial stain, (e.g. CK7, 20 variable) EMA (+) CLA, S-100, Vimentin: (-)
Colorectal carcinoma Lung carcinoma	CK 7 (-); CK 20 (+)
Adenocarcinoma Other	TTF (+), Surf-A ans Surf-B (+)
Non-small cell carcinoma	CK-7 (+), CK 20 (-). TTF-1 (-)
Small cell carcinoma	TTF-1 (+), Chromogranin (+), NSE (+)
Neuroendocrine carcinoma	NSE, Chromogranin, synaptophysin (+) Epithelial stain (+)
Germ cell tumor	HCG, AFP (+) Oct4 transcription factor (+) Placental alkaline phosphatase (+) Epithelial stains (+)
Prostate cancer	PSA (+), Epithelial stains (+) CK-7 (-), CK 20 (-)
Pancreas carcinoma	CA 199 (+), CK 7 (+) Mesothelin (+), trifoil factor 1 (+)
Breast carcinoma	ER, PR (+) Her-2-neu (+) Gross cystic fluid protein 15 (+) Epithelial stains (+) CK-7 (+), CK 20 (-)
Thyroid carcinoma	
Follicular	Thyroglobulin
Medullary	Calcitonin
Lymphoma	CLA(+), rare false (-) EMA: occasionally (+) all other stains (-)
Sarcoma	
Mesenchymal	Vimentin (+), Epithelial stains usually (-)
Rhabdomyosarcoma	Desmin (+)
Angiosarcoma	Factor VIII antigen
Melanoma	S-100, Vimentin (+); HMB-45 (+) NSE often (+) Epithelial stains (-)

拾柒、緊急災害事件動員規定

一、林口院區：

（一）緊急災害動員

1.病房發生緊急災害時，以維護病患及家屬安全為優先，所屬病房醫師應即加入動員，採取撤離及照護等應變，其動員規定如下：

（1）正常上班時間：災害發生部門病患所屬之住院醫師、病房主任及在場之主治醫師。

（2）夜間及假日時段：因夜間人力較少，因此事件發生所屬大樓之各科系值班主治醫師及值班住院醫師（由值班總醫師指派）均應加入救援。

2.急診發生大量傷患（333 紅色）事件時：

（1）外科、骨科、急診醫學科、耳鼻喉科及內科等部、R3（含）以上職級住院醫師及值班照會醫師。

（2）外傷急重症中心及急診醫學科主治醫師。

二、台北院區

院區發生任何緊急災害及急診發生大量傷患（333 紅色）事件，各科值班醫師均應投入救援。

三、為使病房之醫師熟悉相關動員支援之方式，平時醫院辦理演習時應依規定參與動員。（依台北、林口院區緊急災害應變計畫規定）